

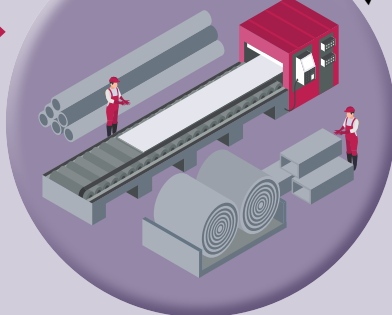
SECTOR SCAN

The steel industry accounts for 10-12% of India's total emissions

India's share in global steel production is 7.4%

It is the largest producer of sponge iron in the world

INDIAN STEEL SECTOR



The country has the fifth-largest iron ore reserve in the world

It is the 2nd-largest producer of crude steel and consumer of finished steel



The cement industry is responsible for 13% carbon emissions

Cement production is set to record a CAGR of 3.1% till 2070

India is the second-largest cement producer in the world, accounting for 8% of total installed capacity

The per capita consumption of cement in India is 257 kg, far less than the global average of 540 kg

mobility. In December 2024, UltraTech Cement, India's largest cement maker, used inland waterways to transport gypsum.

CONSUMPTION TO RISE

The change has come during an opportune period as the per capita steel and cement consumption is set to increase given the massive infrastructure needs of the country. The per capita consumption of cement in India is 257 kg, far less than the global average of 540 kg. And per capita consumption of steel is around 77 kg per year compared with the global average of 233 kg.

"The Ministry of Steel is already working on a green steel public procurement policy. The question is to what extent this policy would support the mandated entities in absorbing the higher costs attached with green steel," says Parth Kumar, industry programme manager at the Centre for Science and Environment. "Mandated preference for low-carbon cement in government tenders could push the demand for low-carbon cement," he says. Soon, public sector tenders may come with a clause for green steel procurement

Brownfield steel plants could cut emissions by up to 40% through hydrogen blending, integrating renewable energy sources and increasing use of scrap, say experts.

Tata Steel is setting up its first electric arc furnace-based plant in Ludhiana, Punjab, to produce green steel. "At our existing blast furnace-based sites, we are leveraging innovation and technology to further decarbonisation. At Jamshedpur, we have pilots like coke oven gas injection, carbon capture and increased scrap use. We're also increasing the share of renewable energy and utilising greener fuels like biochar and natural gas," says Rajiv Mangal, Vice